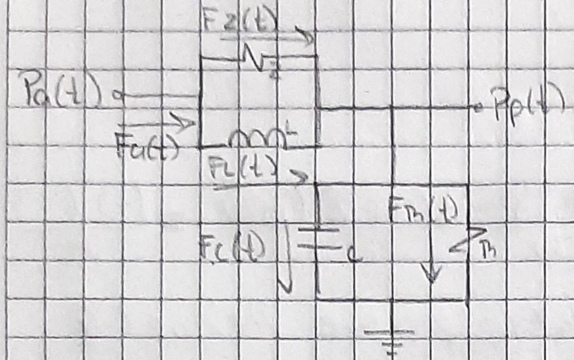
10/10/25

Práctica 5.4: Sistema cardiovascular



① Ecuación principal

$$F_a(t) = F_Z(t) + F_L(t) = F_C(t) + F_R(t)$$

$$F_Z(t) = \frac{P_a(t) - P_p(t)}{Z}$$

$$F_L(t) = \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt$$

$$F_C(t) = C\,dP_p(t) / dt$$

$$F_R(t) = P_p(t) / R$$

② Procedimiento algebraico

$$\frac{P_a(t)}{Z} - \frac{P_p(t)}{Z} + \frac{1}{L}\int [P_a(t) - P_p(t)]\,dt = \frac{C\,dP_p(t)}{dt} + \frac{P_p(t)}{R}$$

$$\frac{P_a(s)}{Z} - \frac{P_p(s)}{Z} + \frac{P_a(s) - P_p(s)}{LS} = CS\,P_p(s) + \frac{P_p(s)}{R}$$

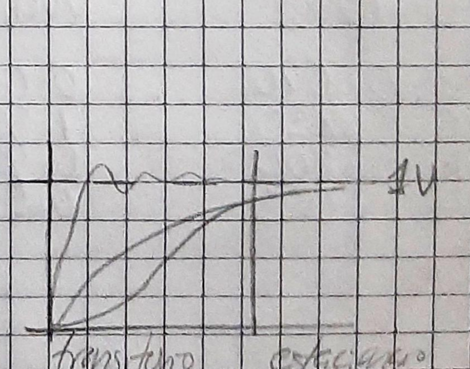
$$\left(\frac{1}{Z} + \frac{1}{LS}\right) P_a(s) = \left(CS + \frac{1}{R} + \frac{1}{Z} + \frac{1}{LS}\right) P_p(s) \rightarrow \frac{1}{RLSZ} + CS$$

③ Función de transferencia

$$\left(\frac{1}{Z} + \frac{1}{LS}\right) \rightarrow \left[\frac{LS + Z}{ZLS}\right] P_a(s) \rightarrow \frac{\frac{LS+Z}{ZLS}}{\frac{RCLS^2 + RLS + LZS + RZ}{RLSZ}}$$

$$= \frac{RLS + RZ}{CLRZS^2 + (LZ + RL)S + RZ}$$

④ Error en estado estacionario



$$e(s) = \lim_{s \to 0} s\,P_a(s)\left[1 - \frac{P_p(s)}{P_a(s)}\right]$$

$$= \lim_{s \to 0} s \cdot \frac{1}{s}\left[1 - \frac{RLS + RZ}{CLRZS^2 + (LZ + RL)S + RZ}\right]$$

$$= 1 - \frac{RZ}{RZ} = 0V$$

Estabilidad en lazo abierto

$$\lambda_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$a = CLBZ$
$b = LZ + BL$
$c = BZ$

$$\lambda_{1,2} = \frac{-(LZ+BL) \pm \sqrt{(LZ+BL)^2 - 4(CLBZ)(BZ)}}{2(CLBZ)} = \frac{(-)(\ )}{+}$$

El sistema tiene una respuesta estable porque $Re\,\lambda_{1,2} < 0$

Modelo de ecuaciones integrodiferenciales

$$P_p(t)\left(\frac{1}{B} + \frac{1}{Z}\right) = \frac{P_a(t)}{Z} + \frac{1}{L}\int\left[P_a(t) - P_p(t)\right]dt - \frac{C\,dP_p(t)}{dt}$$

$$P_p(t) = \left(\frac{P_a(t)}{Z} + \frac{1}{L}\int\left[P_a(t) - P_p(t)\right]dt - \frac{C\,dP_p(t)}{dt}\right)\frac{ZB}{Z+B}$$